

PRAKTIKUM SISTEM OPERASI

MODUL 10

(SHELL)



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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JURNAL/TP

1. Akan dimodifikasi shell dengan modifikasi syscall bernama chname yang berfungsi untuk mengubah nama suatu proses. Lihat kembali modul sebelumnya cara membuat syscall. Perhatikan sekarang syscall chname mempunyai 3 parameter yaitu pid, character dan panjang character. Character untuk menyimpan nama dan panjang character untuk panjang nama.

```
#include <xinu.h>
```

```
syscall chname(pid32 pid, char *newname, uint32 namelen) {
    intmask mask;
    struct procent *prptr;
    int32 i;

    mask = disable();

    if (isbadpid(pid)) {
        restore(mask);
        return SYSERR;
    }

    prptr = &proctab[pid];

    for (i = 0; i < namelen; i++) {
        prptr->prname[i] = newname[i];
    }

    prptr->prname[namelen] = '\0';
    kprintf("New Syscall is easy\n");

    restore(mask);
}
```

C Tab Width: 8 Ln 25, Col 38 INS

```
extern void arp_ntoh(struct arppacket *);
extern void arp_hton(struct arppacket *);

/* in file ascdatetime.c */
extern status ascdatetime(uint32, char *);

/* in file bufinit.c */
extern status bufinit(void);

/* in file chprio.c */
extern syscall chprio(pid32, pri16);

/* in file control.c */
extern syscall chname(pid32 pid, char *newname, uint32 namelen);

/* in file clkupdate.S */
extern uint32 clkcount(void);

/* in file clkhandler.c */
extern interrupt clkhandler(void);

/* in file clkinit.c */
extern void clkinit(void);

/* in file clkdisp.S */
extern void clkdisp(void);

/* in file close.c */
```

C/C++/ObjC Header Tab Width: 8 Ln 28, Col 1 INS

2. Buatlah perintah baru bernama namecmd sesuai dengan langkah-langkah pada no.5 pada modul shell!

```
#include <xinu.h>
#include <stdio.h>
#include <string.h>

shellcmd xsh_namecmd(int nargs, char *args[]) {
    pid32 pid;

    pid = getpid();

    chname(pid, "gantinama", strlen("gantinama"));
    kprintf("Nama proses berhasil diubah\n");

    return 0;
}
```

```
/* in file xsh_namecmd.c */
extern shellcmd xsh_namecmd (int32 nargs, char *args[]);

/* in file xsh_argecho.c */
extern shellcmd xsh_argecho (int32, char *[]);

/* in file xsh_arp.c */
extern shellcmd xsh_arp (int32, char *[]);

/* in file xsh_bingid.c */
extern shellcmd xsh_bingid (int32, char *[]);

/* in file xsh_cat.c */
extern shellcmd xsh_cat (int32, char *[]);

/* in file xsh_clear.c */
extern shellcmd xsh_clear (int32, char *[]);

/* in file xsh_date.c */
extern shellcmd xsh_date (int32, char *[]);

/* in file xsh_devdump.c */
extern shellcmd xsh_devdump (int32, char *[]);

/* in file xsh_echo.c */
extern shellcmd xsh_echo (int32, char *[]);
```

```
/* in file xsh_ethstat.c */
```

```
/* shell.c - shell */

#include <xinu.h>
#include <stdio.h>
#include "shprototypes.h"

/* Table of Xinu shell commands and the function associated with each */
const struct cmdent cmdtab[] = {
    {"namecmd", xsh_namecmd},
    {"argecho", xsh_argecho},
    {"arp", xsh_arp},
    {"cat", xsh_cat},
    {"clear", xsh_clear},
    {"date", xsh_date},
    {"devdump", xsh_devdump},
    {"echo", xsh_echo},
    {"help", xsh_help},
    {"ls", xsh_ls},
    {"kill", xsh_kill},
    {"memdump", xsh_memdump},
    {"memstat", xsh_memstat},
    {"ns", xsh_ns},
    {"netinfo", xsh_netinfo},
    {"ping", xsh_ping},
    {"ps", xsh_ps},
    {"sleep", xsh_sleep}
}
```

3. Testing Hasilnya

```
Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3d
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0d
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0d
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0d
5 shell       recv   50   4 0x005D7FFC 0x005D79AC 8192 0 0d
6 ps         curr   20   5 0x005F7FFC 0x005F7FC4 8192 0 0d
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

```
Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3d
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0d
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0d
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0d
5 shell       recv   50   4 0x005D7FFC 0x005D79AC 8192 0 0d
6 ps         curr   20   5 0x005F7FFC 0x005F7FC4 8192 0 0d
xsh $ namecmd
Nama proses berhasil diubah
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3d
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0d
2 gantinama   wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0d
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0d
5 shell       recv   50   4 0x005D7FFC 0x005D79AC 8192 0 7d
8 ps         curr   20   5 0x005F7FFC 0x005F7FC4 8192 0 0d
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```